STANDARD OPERATING PROCEDURE OF TENSILE FOR TEXTILE VF-B01

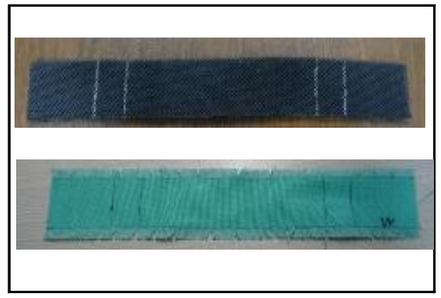
Lab /SOP/ 04-02



1. Tensile testing machine with chart recorder
2. Clamps
3. Steel scale or calipers



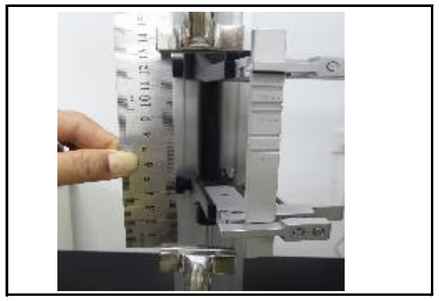
Cut three (3) samples per material in each direction.
non-woven Fabric: 2.54 ± 0.10cm x 20 ± 1cm
Woven Fabric:3.5 ± 0.10cm x 20 ± 1cm
Ravel each specimen to give a test width of 2.54 ± 0.10cm by removing an approximate equal number ofyarns from each side.



Draw two elongation bench marks at 10 cm apart. Measure and record the width of the test sample to the nearest 0.5mm



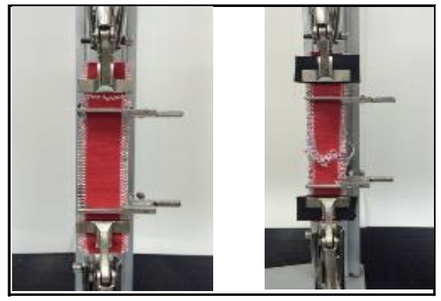
Zero and calibrate the tensile testing machine.



Set the tensile testing machine to the specified load range, crosshead speed and jaw gap.
Crosshead speed =300 mm/minute
Jaw gap =13.0+/- 0.3cm
Elongation bench mark gaps =10.0 +/- 0.1cm



Insert the sample centrally in the top and bottom jaws of the tensile testing machine the sample is vertical.No slack or tension is present
Operate the tensile testing machine at the specified crosshead speed.



Record the breaking load to the nearest 0.1kgf (B)
All samples must break at least 1.5cm above the bottom jaw or 1.5cm below the top jaw, any sample not meeting this criteria is considered a “jaw break”, and must be disregarded and re-tested.



Materials of specified thickness (leather, synthetic leather, fabric etc).
Tensile Strength (kgf/cm) = B / W
Elongation (%) = (D2 - D1) / D1 x 100
Tensile strength is reported in kgf/cm.

Prepared by	Approved by	REVISION		DATE	